

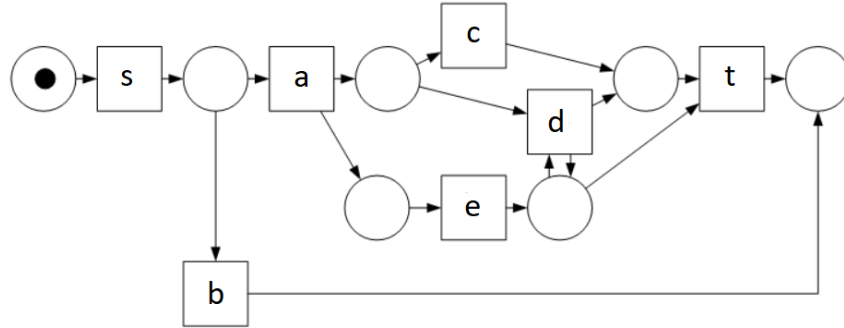
PODS EXAM: Practical Part

Name:	
Student ID:	

Exercise 0.1 The following event log was extracted from an enterprise information system $L = [\langle a, d, f, h \rangle, \langle a, c, e, g, c, e, h \rangle, \langle b, f, g, d, f, h \rangle]$

- Derive the $\rightarrow_L$ relation.
- Use the eight steps of the α -algorithm to construct the corresponding Petri net and draw the Petri net (delivering all of the intermediate results is not necessary, only the resulting Petri net is required).
- If possible, give a trace possible according to the discovered model but not (yet) observed in the log.

Exercise 0.2 Consider event log $L = [\langle s, a, c, e, t \rangle^5, \langle a, e, d, t \rangle^3, \langle s, a, e, t \rangle^4, \langle s, b \rangle^{10}]$ and the following Petri net:



1. Compute fitness of the event log with respect to the model using token-based fitness.
2. Give an optimal alignment for the trace $\langle a, b, d, c, d, e, t \rangle$, and, its corresponding fitness. Use the standard cost function, i.e. log/model moves have cost 1, synchronous moves have value 0.

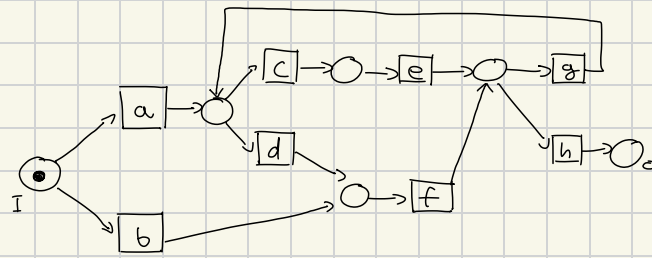
Exercise 0.1 The following event log was extracted from an enterprise information system $L = [\langle a, d, f, h \rangle, \langle a, c, e, g, c, e, h \rangle, \langle b, f, g, d, f, h \rangle]$

- Derive the $\rightarrow_L$ relation.
- Use the eight steps of the α -algorithm to construct the corresponding Petri net and draw the Petri net (delivering all of the intermediate results is not necessary, only the resulting Petri net is required).
- If possible, give a trace possible according to the discovered model but not (yet) observed in the log.

	a	b	c	d	e	f	g	h
a	#	#	$\rightarrow$	$\rightarrow$	#	#	#	#
b	#	#	#	#	#	$\rightarrow$	#	#
c	$\leftarrow$	#	#	#	$\rightarrow$	#	$\leftarrow$	#
d	$\leftarrow$	#	#	#	#	$\rightarrow$	$\leftarrow$	#
e	#	#	$\leftarrow$	#	#	#	$\rightarrow$	$\rightarrow$
f	#	$\leftarrow$	#	$\leftarrow$	#	#	$\rightarrow$	$\rightarrow$
g	#	#	#	$\rightarrow$	$\rightarrow$	$\leftarrow$	#	#
h	#	#	#	#	$\leftarrow$	$\leftarrow$	#	#

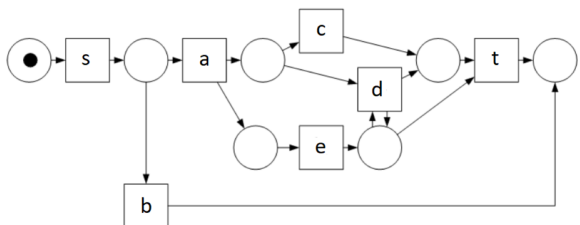
$$\textcircled{1} \rightarrow_L = \{ \langle \cancel{a}, \cancel{d} \rangle \langle \cancel{a}, \cancel{d} \rangle \langle \cancel{b}, \cancel{f} \rangle \langle c, e \rangle \langle \cancel{d}, \cancel{f} \rangle \langle \cancel{e}, \cancel{g} \rangle \langle \cancel{e}, \cancel{h} \rangle \langle \cancel{f}, \cancel{g} \rangle \langle \cancel{f}, \cancel{h} \rangle \langle g, c \rangle \langle g, d \rangle \}$$

$$\textcircled{2} = \{ \langle a, c, d \rangle \langle (b, d), f \rangle \langle c, e \rangle \langle e, g, h \rangle \langle f, g, h \rangle \langle g, c, d \rangle \}$$



$$\textcircled{3} \langle g, f, h \rangle$$

Exercise 0.2 Consider event log $L = [\langle s, a, c, e, t \rangle^5, \langle a, e, d, t \rangle^3, \langle s, a, e, t \rangle^4, \langle s, b \rangle^{10}]$ and the following Petri net:



1. Compute fitness of the event log with respect to the model using token-based fitness.
2. Give an optimal alignment for the trace $\langle a, b, d, c, d, e, t \rangle$, and, its corresponding fitness. Use the standard cost function, i.e. log/model moves have cost 1, synchronous moves have value 0.

$\textcircled{1}$	Θ_1	Θ_2	Θ_3	Θ_4	
M	= 0	1	= 3	1	= 4
C	1+1+1+1+2+1=35	1+1+2+2+1=21	1+1+1+2+1=24	1+1+1=30	105
P	1+1+2+1+1+1=35	1+2+1+2+1=21	1+1+2+1+1=24	1+1+1=30	105
R	= 0	1	= 3	1	= 4

$$\text{Fitness} = \frac{1}{2} \left(1 - \frac{M}{C} \right) + \frac{1}{2} \left(1 - \frac{R}{P} \right) = 1 - \frac{7}{105} = \frac{98}{105}$$

$\textcircled{2}$	L_{log}	$\gg a \ b \ d \ c \ d \ e \ t$	$\text{Cost} = 4$	$\text{Fitness} = 1 - \frac{\text{Cost}}{ \text{Trace} + \text{Shortest path} } = 1 - \frac{4}{7+2} = \frac{5}{9}$
	Model	$\gg s \ a \gg \gg c \gg e \ t$		